

Advanced (Habanero)

- Q1.** Classify the number 5 as?
(A) Natural and whole
(B) Integer and whole
(C) Natural, whole, and integer
(D) Only natural
(E) Only whole
- Q2.** Which of the following is an example of an even number?
(A) 1
(B) 3
(C) 7
(D) 10
(E) 11
- Q3.** The pattern 2, 5, 8, 11, 14, ... is an example of?
(A) An arithmetic sequence with a common difference of 3
(B) A geometric sequence with a common ratio of 2
(C) An arithmetic sequence with a common difference of 2
(D) A geometric sequence with a common ratio of 3
(E) Neither arithmetic nor geometric
- Q4.** Evaluate the expression $(-3) \cdot (-2)$
(A) -6
(B) 6
(C) -1
(D) 1
(E) 0
- Q5.** Simplify $4 + (-2)$
(A) 2
(B) 6
(C) -2
(D) -6
(E) 0
- Q6.** What is the value of x in the equation $x - 2 = 7$?
(A) 5
(B) 9
(C) 3
(D) 1
(E) 0
- Q7.** Which of the following numbers is a whole number but not a natural number?
(A) 0
(B) 1
(C) -1
(D) 2
(E) -2
- Q8.** The next number in the sequence 1, -2, 3, -4, 5, ... is?
(A) -6
(B) 6
(C) -5
(D) -3
(E) 7

Open Ended Questions (FRQ)

Q9. Prove that the sum of any two odd integers is even

Q10. Find the next three numbers in the pattern 1, 4, 7, 10, 13, ... and explain your reasoning

Chef's Tips

- Emphasize the difference between natural and whole numbers
- Use visual aids to demonstrate patterns and sequences
- Encourage students to use mathematical notation and vocabulary